

Goutham Arelli

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SUMMARY

Backend Engineer with 4+ years of experience building scalable microservices, distributed backend systems, and reliability-focused platform features. Production experience spans Kotlin, Java, Spring Boot, Node.js, MySQL, caching, observability, and high-throughput EMI workflows. Recently applying this backend foundation to AI-agent infrastructure through MCP and agentic commerce prototypes, using Codex-assisted development to accelerate implementation while owning system design, workflows, and integration behavior.

SKILLS

- **Languages:** Java, Kotlin, TypeScript, JavaScript, SQL
- **Backend:** Spring Boot, Node.js, Express.js, Fastify, REST APIs, Microservices, API Gateway Design
- **AI / Agent Infrastructure:** Model Context Protocol, MCP Servers, Agentic Commerce, Tool Permissions, Connector Concepts, Provider Onboarding Flows, AI-Assisted Prototyping
- **Scale / Reliability:** Kotlin Coroutines, Caching, Circuit Breakers, Retry Policies, Timeout Management, OpenTelemetry, Distributed Tracing
- **Databases / Frontend:** MySQL, PostgreSQL, Redis, Angular, React, Next.js

EXPERIENCE

Backend Engineer

May 2025 - Present

Pine Labs

- Designed and scaled **distributed microservices** for EMI workflows, improving throughput, fault tolerance, and reliability across high-volume payment systems.
- Increased **EMI Offer Discovery capacity by 2000+ TPS** through Kotlin coroutine-based concurrency optimization, caching improvements, and refined content negotiation plugins.
- Improved production **observability** by integrating OpenTelemetry Java Agent and adding custom span attributes for granular trace filtering, metric analysis, and service diagnostics.
- Built a modular **HTTP Client Factory** using the Singleton pattern, enabling dedicated service clients with configurable timeouts, retries, and circuit breaker controls.
- Contributed to an **agentic commerce framework** that allows merchants to expose catalog, cart, checkout, payments, and order workflows through a common protocol.
- Enabled **AI-agent-compatible commerce flows** where buyer and seller agents can coordinate product discovery, checkout, and payment interactions across providers.

FullStack Engineer

Jan 2024 - Jul 2024

Eduspeed Technologies

- Built and enhanced an **EdTech platform** using Angular and Spring Boot, improving performance, scalability, and product functionality across coaching center and assessment workflows.
- Led development of UI components and backend services for coaching center management and assessment modules from the ground up.
- Reduced backend calls by **50%** through caching, lazy loading, and resource-saving frontend/backend interaction patterns.
- Designed optimized database schemas and bulk update mechanisms, improving large dataset handling efficiency by **30%**.

Software Development Engineer (Full Stack)

Aug 2021 - Jan 2024

Cognizant

- Designed and developed a full-stack telecom solution that centralized customer data, reducing query resolution time by **40%** and improving service efficiency.
- Architected scalable Spring Boot microservices using MVC design and SOLID principles for maintainable, extensible architecture.
- Engineered a MySQL data layer with optimized schema design for efficient processing of large-scale telecom customer data.
- Implemented secure authentication with JWT, Eureka Server, and API Gateway, while building a responsive Angular frontend that improved user satisfaction by **20%** and scalability by **35%**.

PROJECTS

MCP Nexus

AI-assisted prototype for a secure MCP gateway and connector marketplace.

- Built an **AI-assisted TypeScript monorepo prototype** for a Model Context Protocol gateway that explores how users can connect once and access approved third-party MCP providers through a managed connector layer.
- Designed product and system architecture for customer, admin, and developer portal flows, including connector discovery, provider onboarding, MCP server submissions, validation jobs, tool discovery previews, and admin approval gates.
- Prototyped backend and frontend modules using Fastify, Next.js, React, PostgreSQL, Redis, BullMQ, Zod, Vitest, MCP SDK, and pnpm workspaces.
- Explored governance requirements for MCP platforms, including user authentication, role-based access, provider token boundaries, per-tool permissions, audit logging, rate limits, emergency kill switches, and high-risk tool confirmation flows.
- Used **Codex-assisted development** to convert architecture, workflow, and product requirements into working platform modules while reviewing generated code for system behavior and integration fit.

ACHIEVEMENTS

- Received H1 FY 25-26 Excellence Award at Pine Labs Technologies.
- Received Star Performer of the Quarter at Eduspeed Technologies.

EDUCATION

Scaler Neovarsity

2025

MS in Computer Science

VIT, Vellore

2021

BE/B.Tech/BS in ECE in IoT | 8.46 CGPA